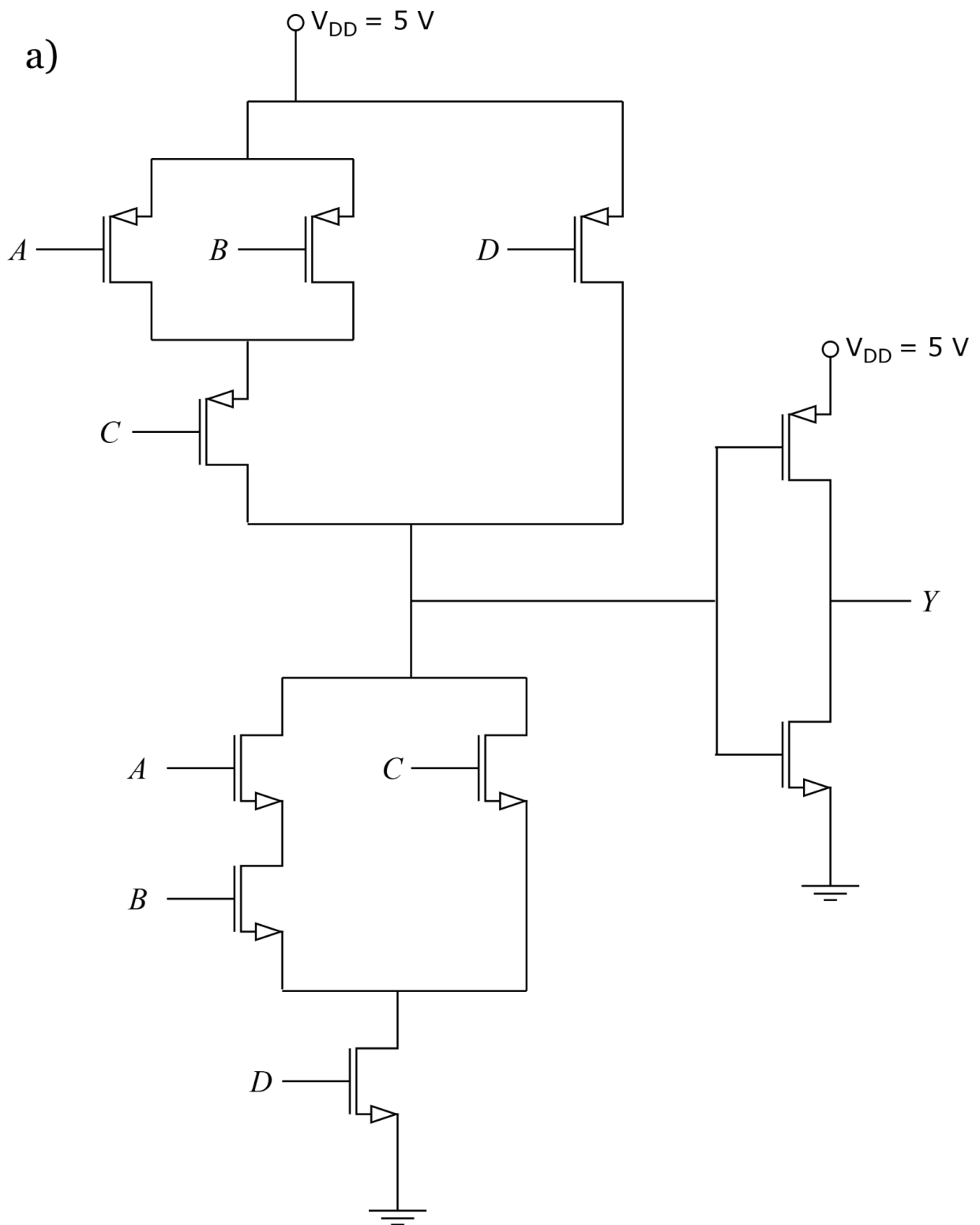


BRAC UNIVERSITY
Department of Computer Science & Engineering
Practice Problem Sheet - CMOS Logic
CSE 350: Digital Electronics and Pulse Technique

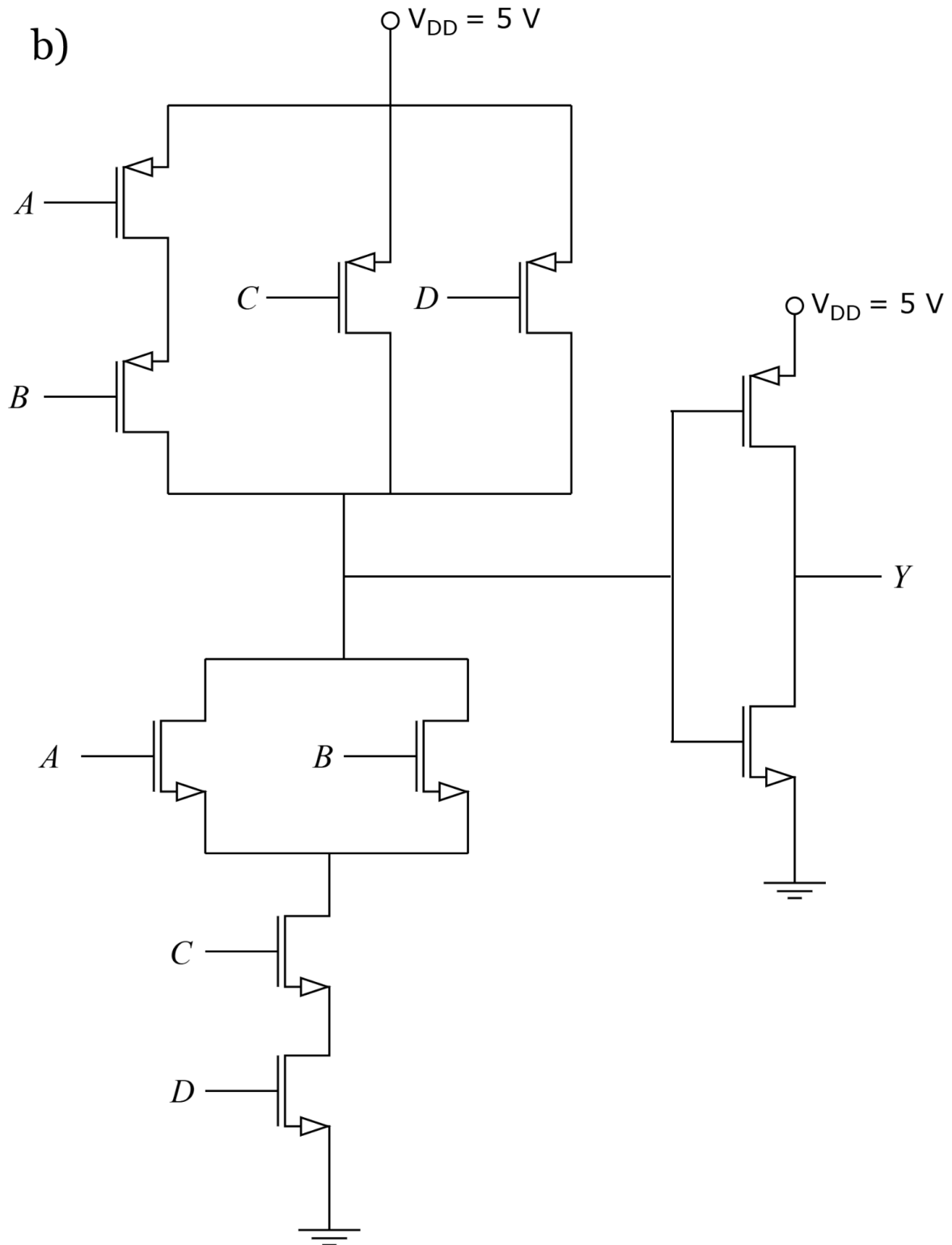
Question 1

(a)	Design a static CMOS logic circuit that implements the logic function $Y = (AB+C)D$
(b)	Design a static CMOS logic circuit that implements the logic function $Y = (A+B)CD$
(c)	Design a static CMOS logic circuit that implements the logic function $Y = F(AB+C(D+E))$
(d)	Design a static CMOS logic circuit that implements the logic function $Y = \overline{ABC + DE}$
(e)	Design a static CMOS logic circuit that implements the logic function $Y = \overline{A} + B$

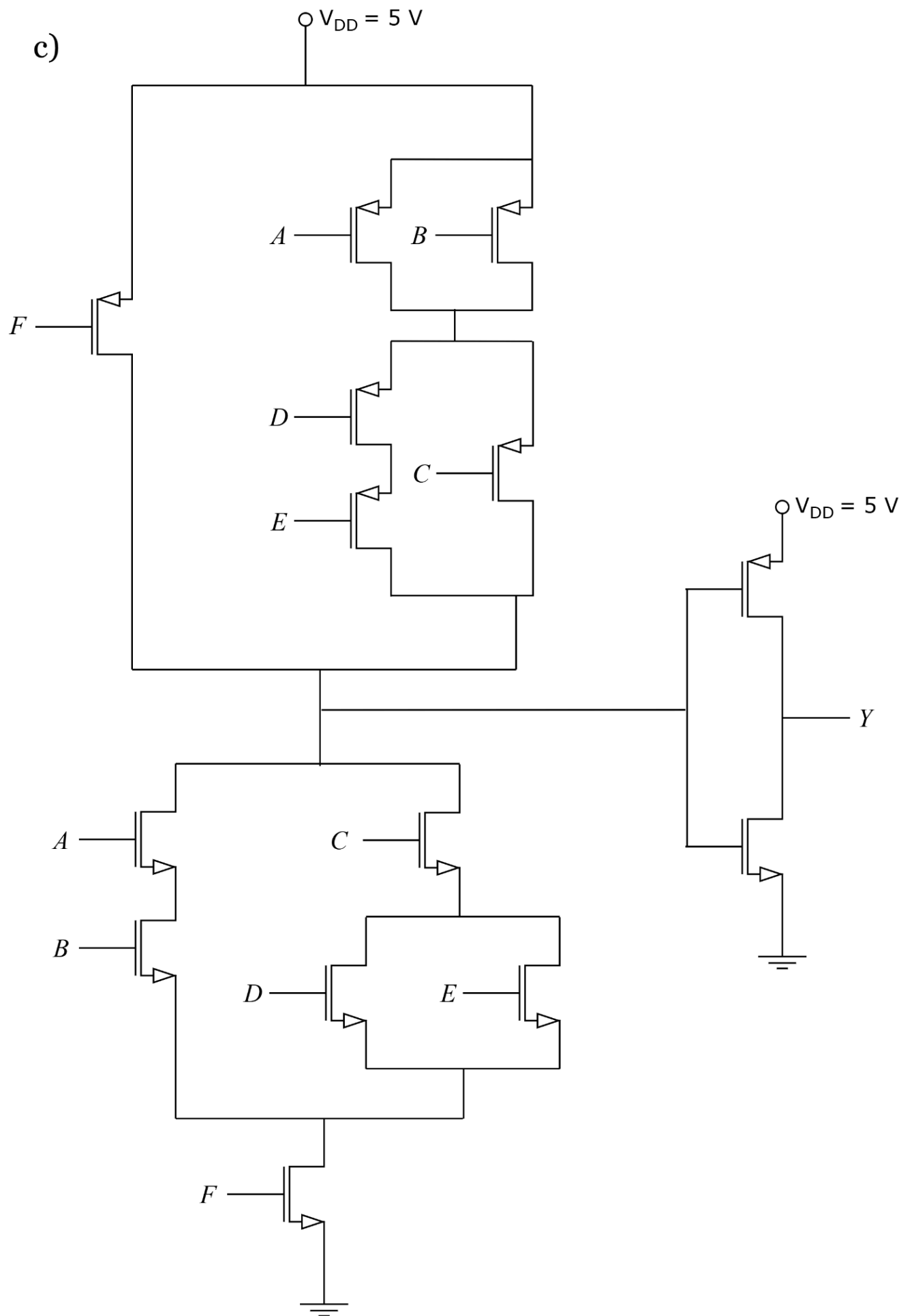
Solution:



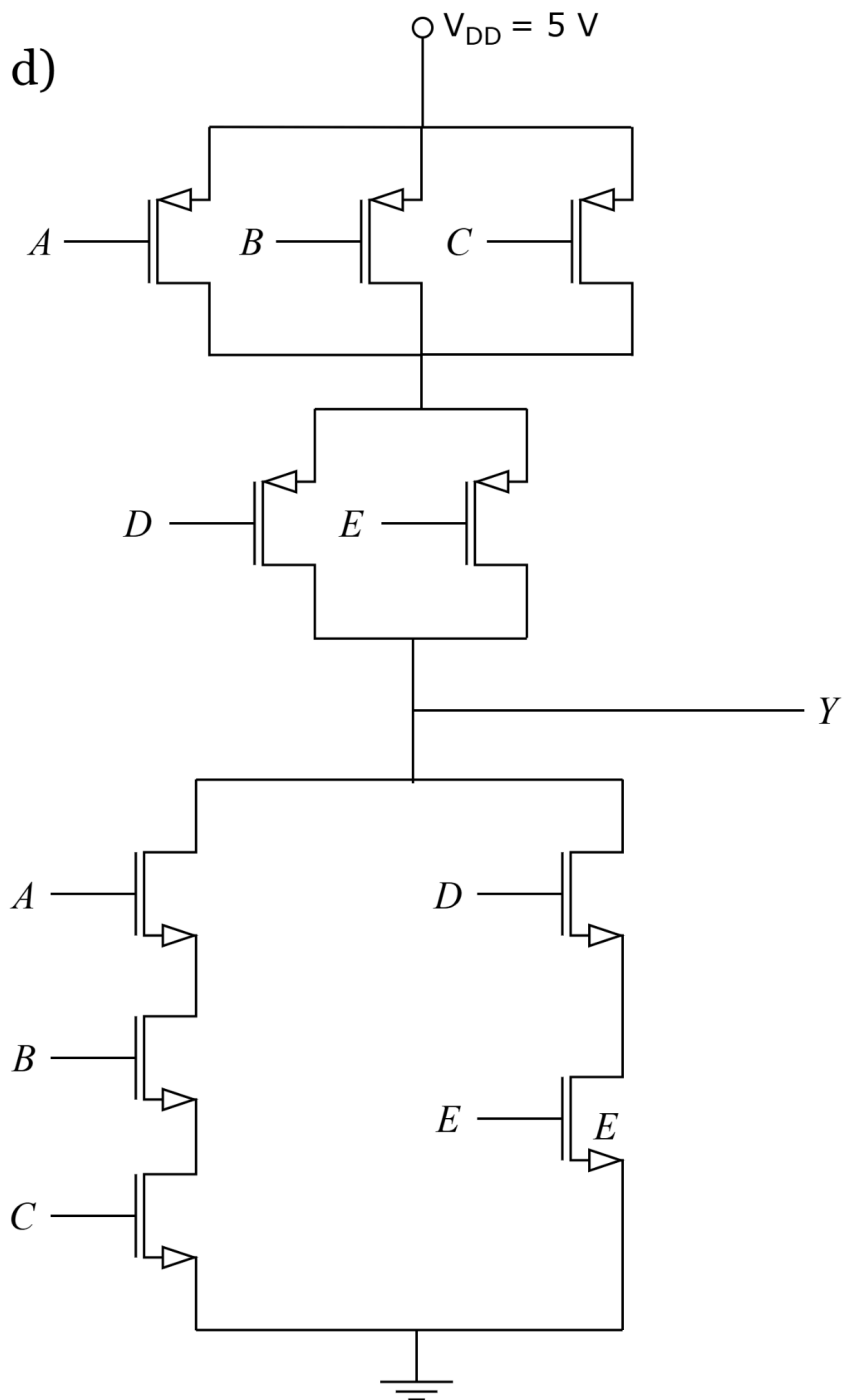
b)



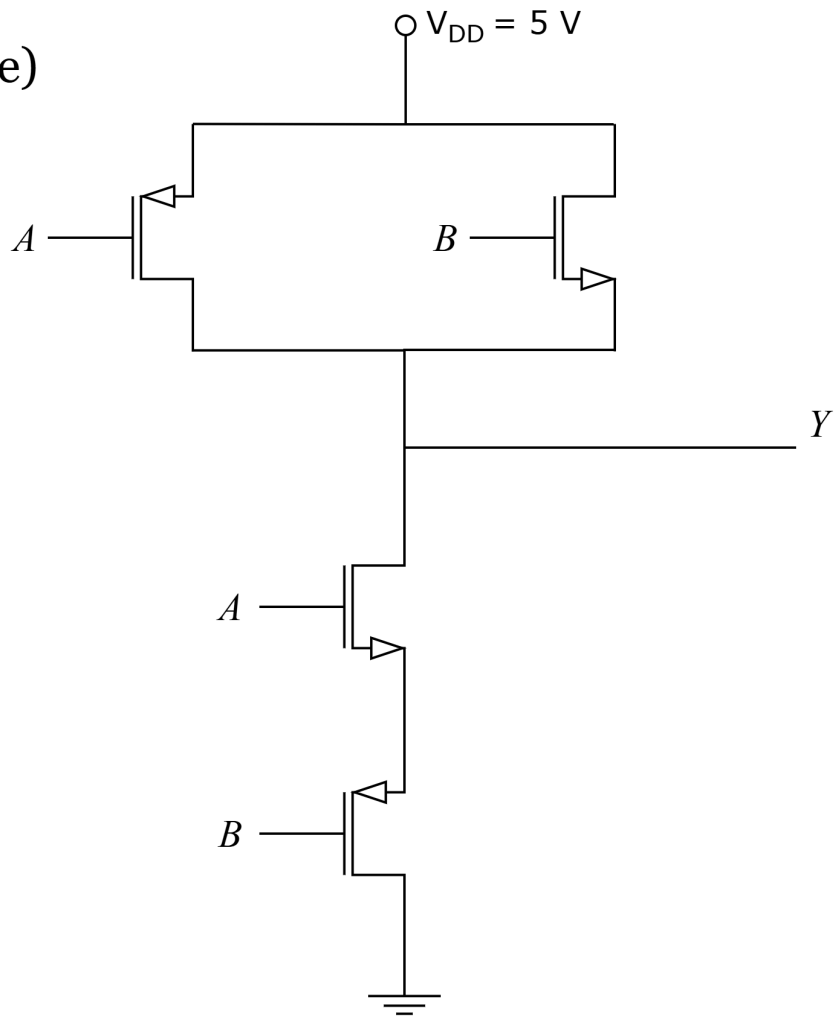
c)



d)

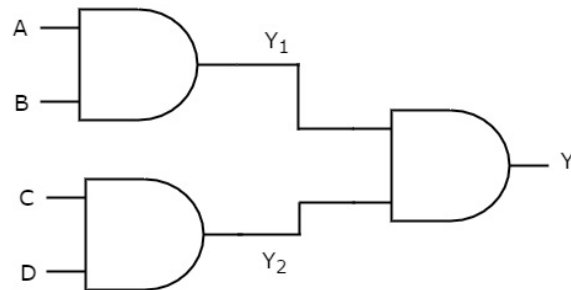


e)



Question 2

Design a CMOS logic circuit to implement the given compound gate in Figure below. First derive the logical expression of output Y and then design the CMOS network.



Solution:

$$\begin{aligned} Y &= Y_1 \cdot Y_2 \\ &= (A \cdot B) \cdot (C \cdot D) \\ &= A \cdot B \cdot C \cdot D \end{aligned}$$

